Make 11

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Track A: Carbon Footprint Visualization

Visualize the CO_2 cost of AI training and use.

Data to find:
- GPT-3 training: ~552 tons CO_2
- One ChatGPT query: ~4.55g CO_2 (estimate)
- Average US household: ~48 tons CO_2/year
- Flight NYC-London: ~986 kg CO_2

Make it visceral:
- How many flights = one AI training run?
- How many trees needed to offset?
- How many years of personal use?

1. Environmental Visualization

- Data visualization, map, infographic, or artwork
- Clear data sources
- Makes the abstract (carbon, water) visceral

AI & Sustainability

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1 TREE

needs to be planted for every ~10,000 AI queries to offset CO_2 emissions

ChatGPT takes

10x

the energy of a google search

Data centers use 4% of the United States' energy

2. Research Documentation (300 words)

- What sources did you use?
- What data was hardest to find?
- What uncertainties exist?

Something really interesting I noticed immediately was that if you ask ChatGPT for any of this information, all of its answers are skewed to be biased towards AI. Aside from the statistics, it kept saying things like, “A **single AI query is tiny** compared to daily activities… about the same as a few Google searches or seconds of video streaming” to try to minimize it. Additionally, it said, “**Training a model is expensive—but rare…**it’s a one-time hit spread across millions (or billions) of uses,” to try to show that it affects millions of people with one training, again, trying to minimize the impacts. It made sure to point out that “**Transportation dominates**…one long-haul flight = *hundreds of thousands* of AI queries” which shows that it once again is trying to show that there are other things that people do often that are more environmentally harmful. Something I found really interesting was that data centers consume more than 4% of U.S. energy, and that this is equal to 80,000 homes’ worth of energy (CBS video). This is such a staggering number to me. These data centers at the current moment are not sustainable, and using renewable energy sources is really important for future considerations. I also learned that ten times more electricity is used for a ChatGPT search than a regular google search (CBS video). That makes us really need to consider what we are using ChatGPT for, and whether a google search could do the same thing. This is because google searches take much less energy. Some uncertainties are whether renewable energy is in the plans of these AI companies to be used to cool the data centers, because this would be a necessary change to make things more sustainable. I wonder whether these companies will be willing to do this. In terms of sources, I tried to use reputable sources such as CBS News, but also I used ChatGPT to translate some of the quantities to each other, which is when I ran into the bias towards AI use that was given to me by ChatGPT.

3. "Sustainability & Ethics" Web Page

- Live on your course site
- All required sections
- Honest personal reflection

4. Artist Statement (250 words)

- What environmental cost most concerns you?
- Can AI be sustainable? What would it take?
- What is your responsibility?

Perhaps the most important thing I noticed was the comparison of trees needed to be planted to offset the CO2 emissions, because it really quantifies it for me in terms of nature. We need to respect our planet and treat it well. Also, thinking about it in terms of large expenditures helps put it into context too. I think that right now, AI is not sustainable at all, but it definitely could be. The data centers use 4% of the energy of the U.S., which is a large amount for one singular industry, let alone one type of building. This shows how important it is to switch to a renewable energy source for these data centers because AI is becoming more popular. This could be done by using solar or wind energy to cool these plants, because these are much more sustainable forms of energy. If this is able to be done, the carbon footprint may be lighter after these changes are made.I think after learning that ChatGPT takes ten times the energy of a google search, I will be trying to use google more than ChatGPT (as I already do for the most part) to try to do my part. It makes me sad that the environment could be hurt by our actions, and I already try to recycle and not leave water running in my life to help the planet, so this is another version of that in a way. I will also try to convince my friends to do the same. I hope that people will be able to see how important it is to be responsible with our AI use, for many reasons.